

[illegible]

$$Y_1 = 3 \binom{(4)}{\kappa_1} + 5 \binom{(4)}{\kappa_{15}} \text{ \& } Y_2 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3} \text{ \& } \text{Det}(0^+) = -\frac{1}{2} k^2 (3 \binom{(4)}{\kappa_1} + 5 \binom{(4)}{\kappa_{15}}) \text{ \& } \text{Det}(1^+) = \frac{3}{2} k^2 (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3})$$
$$\begin{aligned} & \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^X \mathcal{K}^\alpha_{\alpha}{}^\beta + \frac{5}{3} \overset{(4)}{\kappa}_{15} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^X \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \\ & 6 \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 2 \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + 3 \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \\ & 4 \overset{(4)}{\kappa}_{15} \partial^X \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$
Resolved unitarity condition(s):

(Demonstrably impossible)